

Applied Optimization

Exercise 9-Solution

Equality Constrained Optimization II

Problem 1: Constraints Elimination (0.5 pt)

Solve the following non-linear optimization problem in $\mathbb{R}^3$ with lineal equality constraints by constraint elimination:

$$\begin{aligned} & \text{minimize } f(x) = \exp(\mathbf{1}^T x) + \exp(-\mathbf{1}^T x) \\ & \text{subject to: } \begin{pmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \end{pmatrix} x = \begin{pmatrix} 0 \\ 5 \end{pmatrix} \end{aligned}$$

Hint: use Gaussian elimination or QR decomposition. You are allowed to use a computer algebra tool to verify your computations.

Solution:

By gauss elimination, we start by adding an extra row to the constraints $Ax = b$.

$$\begin{aligned} \begin{pmatrix} 1 & -2 & 0 & | & 0 \\ 2 & 1 & 0 & | & 5 \\ 0 & 0 & 1 & | & z \end{pmatrix} & \xrightarrow{E_2 - 2E_1} \begin{pmatrix} 1 & -2 & 0 & | & 0 \\ 0 & 5 & 0 & | & 5 \\ 0 & 0 & 1 & | & z \end{pmatrix} \xrightarrow{E_2/5} \begin{pmatrix} 1 & -2 & 0 & | & 0 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & z \end{pmatrix} \\ & \xrightarrow{E_1 + 2E_2} \begin{pmatrix} 1 & 0 & 0 & | & 2 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & z \end{pmatrix} \end{aligned}$$

Thus we can write now $\hat{f}(z) = f(Fz + x_0)$ with

$$x_0 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \text{ and } F = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

to obtain

$$\hat{f}(z) = f(2, 1, z) = \exp(3 + z) + \exp(-3 - z)$$

taking derivative with respect to z and setting to 0 we obtain

$$\hat{f}'(z) = e^3 e^{+z} - e^{-3} e^{-z} \stackrel{!}{=} 0$$

$$e^3 e^{+z} = e^{-3} e^{-z}$$

$$e^{+z} = e^{-6} e^{-z}$$

$$e^{+2z} = e^{-6}$$

$$2z = -6$$

$$z = -3$$

Then the solution is $x^* = (2, 1, -3)^T$

Problem 2: Solving the KKT system (0.5 pt)

Solve the following convex quadratic problem in $\mathbb{R}^3$ with lineal equality constraints

$$\begin{aligned} &\text{minimize } f(x) = \begin{pmatrix} -3 & -2 & 7 \end{pmatrix} x + \frac{1}{2} x^T \begin{pmatrix} 6 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 6 \end{pmatrix} x \\ &\text{subject to: } \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \end{aligned}$$

Hint: use gaussian elimination or QR decomposition to solve the resulting KKT system.

Solution:

We start by writing the KKT system

$$\begin{pmatrix} H & A^T \\ A & \mathbf{0} \end{pmatrix} \begin{pmatrix} x \\ \nu \end{pmatrix} = \begin{pmatrix} -q \\ b \end{pmatrix}$$

with

$$H = \begin{pmatrix} 6 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 6 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}, \quad q = \begin{pmatrix} -3 \\ -2 \\ 7 \end{pmatrix} \text{ and } b = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

Then we solve the system with the help of an algebra system, in this case octave version 5.1.2 give is: (\ left division calls the appropriate linear solver for us)

```
> M = [6 0 2 1 0; 0 4 0 1 1; 2 0 6 2 1; 1 1 2 0 0; 0 1 1 0 0]
M =
```

```

6    0    2    1    0
0    4    0    1    1
2    0    6    2    1
1    1    2    0    0
0    1    1    0    0
```

```
> rhs = [3 2 -7 0 2]';
rhs =
```

```

3
2
-7
0
2
```

```
> sln = M\rhs
sln =
```

```

-1.0000
 3.0000
-1.0000
11.0000
-21.0000
```

a simple verification is performed

```
> M*sln
ans =
```

```

3.0000e+00
2.0000e+00
-7.0000e+00
-4.4409e-16
2.0000e+00
```